

AI Engineering Summer Training

Daily Task Submission

Topic: Project Reflection (Week 2 - Day 5)

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Project Reflection

Where AI Helped:

- **Rapid Prototyping & Boilerplate:** AI significantly accelerated the initial development by generating the foundational `Gradio` UI code and the Python `Ollama` API integration.
- **UI/UX Styling:** AI efficiently generated modern CSS styling (dark mode, hovering effects, premium buttons) that transformed a basic script into a professional, cybersecurity-themed SaaS interface.
- **Syntax & Error Handling:** It successfully identified missing library imports, structured the `try-except` blocks for the local server connection, and isolated layout issues by grouping questions into modern UI cards.

Where Manual Intervention was Necessary:

- **Logical Grading Bugs:** The AI blindly followed prompt instructions literally (e.g., hardcoding "Option A" instead of the actual answer text in the JSON), which completely broke the grading system. I had to manually engineer and refine the system prompt constraints to fix this.
- **Architectural & Server Issues:** The AI suggested using `0.0.0.0` for the `Gradio` launch, which caused an `ERR_ADDRESS_INVALID` on the local Windows environment. I intervened to revert it to standard `localhost` settings.
- **UI Component Ordering:** Despite clear instructions, the AI repeatedly struggled with spatial layout logic (placing the 'Submit' button above the generated questions). Manual code review and strict constraints were required to force the correct procedural flow in `Gradio`.